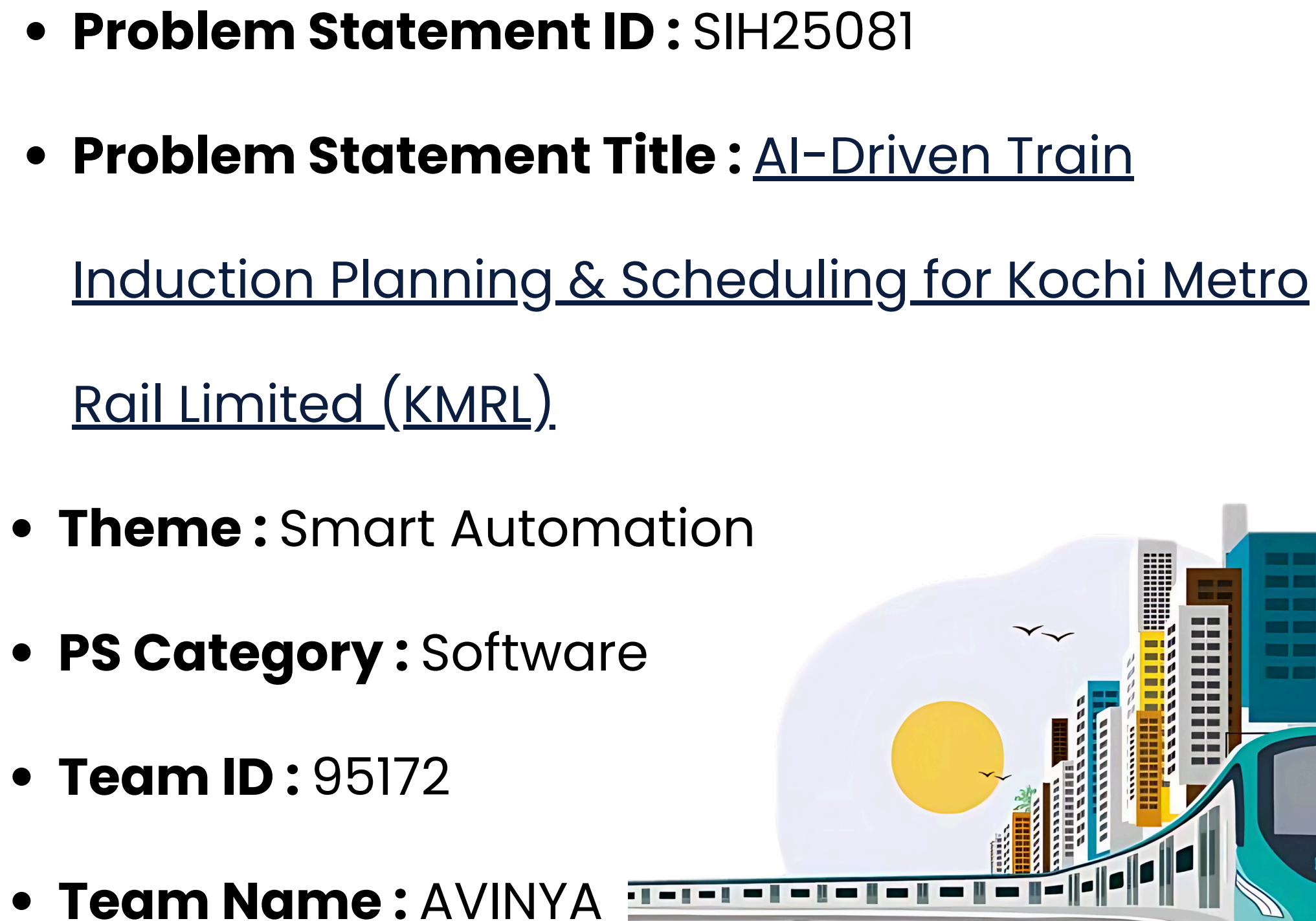


SMART INDIA
HACKATHON
2025



Proposed Solution

Problems Addressed

1

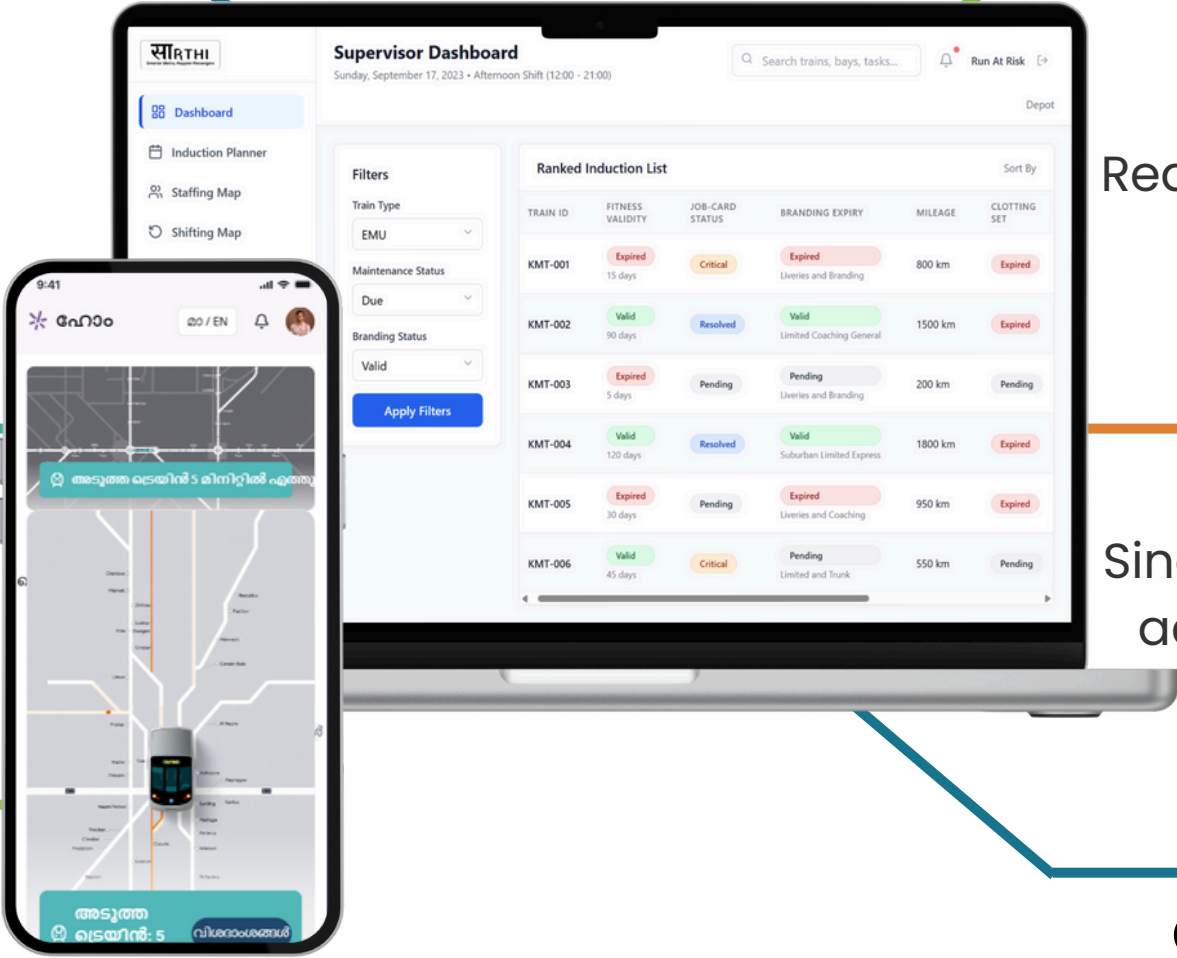
AI-Driven Train Induction
Smart allocation of trains for service, standby, maintenance

2

Dashboard Visualization
Clear status on fitness, mileage, cleaning, branding, stabling.

3

Agentic AI Assistant
Provides suggestions, conflict alerts & what-if simulations



4

Passenger Features
Real-time GPS, complaint logging, multi-language transcription

5

Mobile-Based Platform
Single app with secure role-based access (supervisor + passenger)

6

Optimized and Scalable
Lightweight, future-ready, supports fleet growth & multi-depots

Error-prone data reconciliation → Integrate all disparate data sources : spreadsheets, sensors, Maximo with ETL automation

Excessive shunting and delays → Optimize stabling and bay positions using rules and algorithms

Uneven mileage distribution → Use smart algorithms to balance kilometres

Branding SLA violations → Track and schedule rakes to meet external wrap exposure requirements

Scalability as fleet grows → AI & ML feedback loop adapts to increased trainsets and depots

1) Hybrid data fusion -

- Structured – Maximo
- Semi-structured – Certifications
- IoT – Sensor & GPS data

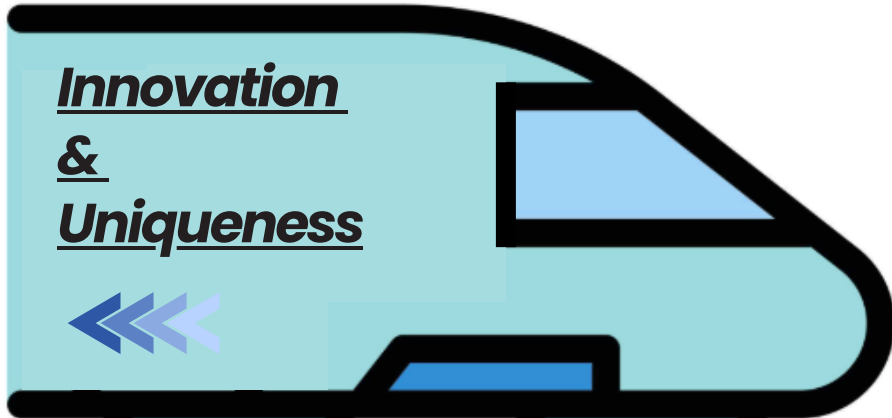
2) LLM-powered reasoning -

- Natural language recommendations
- Conflict alerts for easy supervisor adoption

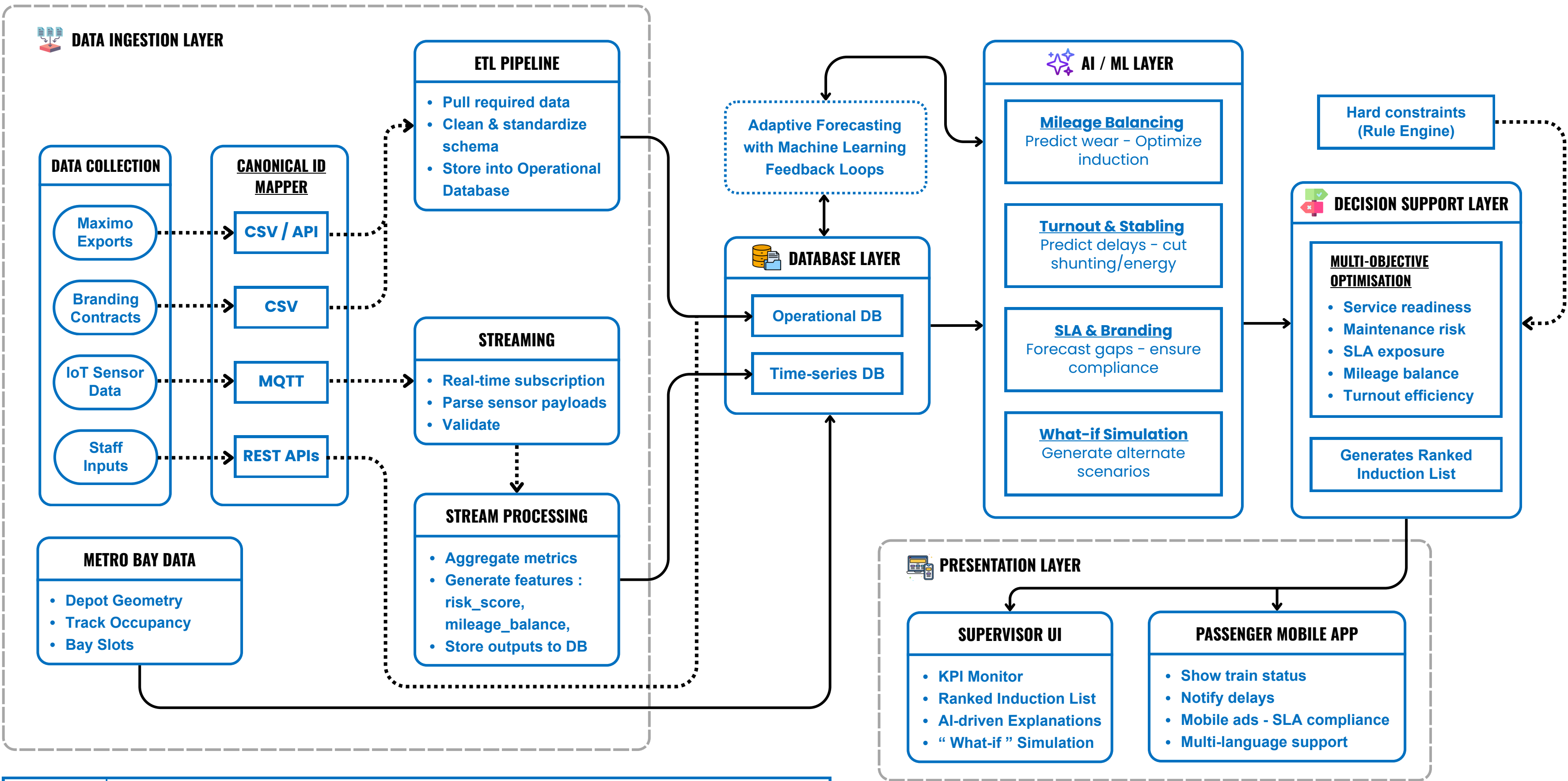
3) Interactive simulations -

- “What-if” simulations to test scenarios & KPI impacts*
- Minimize downtime risk with proactive insights*

Innovation & Uniqueness



TECHNICAL APPROACH



TECHNOLOGY USED

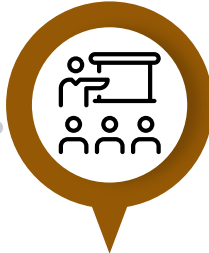


Feasibility



Technical Feasibility

- Integrates Maximo, IoT, and manual overrides.
- Supports 25–40+ trains.



Operational Efficiency

- Maximo data, maintenance logs, mileage, cleaning & stabling info



Economic Feasibility

- Saves ₹10–15 lakh/month,
- 15–20% O&M reduction.



Multilingual Support

- Achievable using existing NLP Transcription APIs

Viability



User-Centric Features

- GPS, complaints, and multilingual support boost public trust and adoption.



Ease of Adoption

- LLM reasoning & what-if sims ease supervisor use.



Data Security

- Encrypted, role-based, auditable access.



Future Scalability

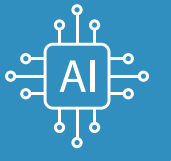
- Ready for 40+ trains and multi-depots by 2027.

Challenges & Mitigation



Incomplete / Inconsistent data

Data Validation and Cleaning



Wrong AI Ranking

Testing with history + What if sims



Lower User Adoption

Simple UI + Training



Fleet Expansion

Modular scalable Backend



GPS Connectivity

Reliable API's + Fallback



Multi - language gaps

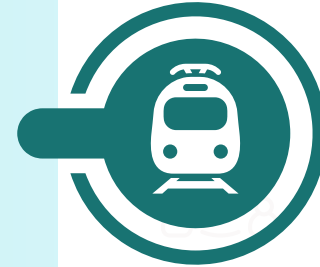
NLP Based Translation

IMPACT AND BENEFITS

Passenger Impact



- **Real-time Train Tracking** → GPS-based updates for accurate train location
- **Inclusive Communication** → Multilingual transcription for all passengers
- **Transparent Issue Resolution** → Easy complaint logging and tracking



Social Benefits



- **Inclusive Access** → Multilingual support and easy complaint logging
- **Enhanced Safety** → Fewer unscheduled rake withdrawals ; safer tracks
- **Smart Mobility** → Supports sustainable urban transportation growth



Operational Impact



- **Automated Train Induction** → Fewer human errors
- **Smart Scheduling** → Optimized standby & maintenance
- **Faster Night Planning** → Reduces workload



Economic Benefits



- **Lower Costs** → Optimized mileage
- **Revenue Protection** → Meets branding commitments
- **Higher Availability** → More trains → more ticket sales



Technological Impact



- **Agentic AI** → "What-if" simulations & intelligent suggestions
- **Data Visualization Dashboard** → Transparent & data-driven decisions
- **Scalable Solution** → Adaptable to any metro network across India



Community Benefits



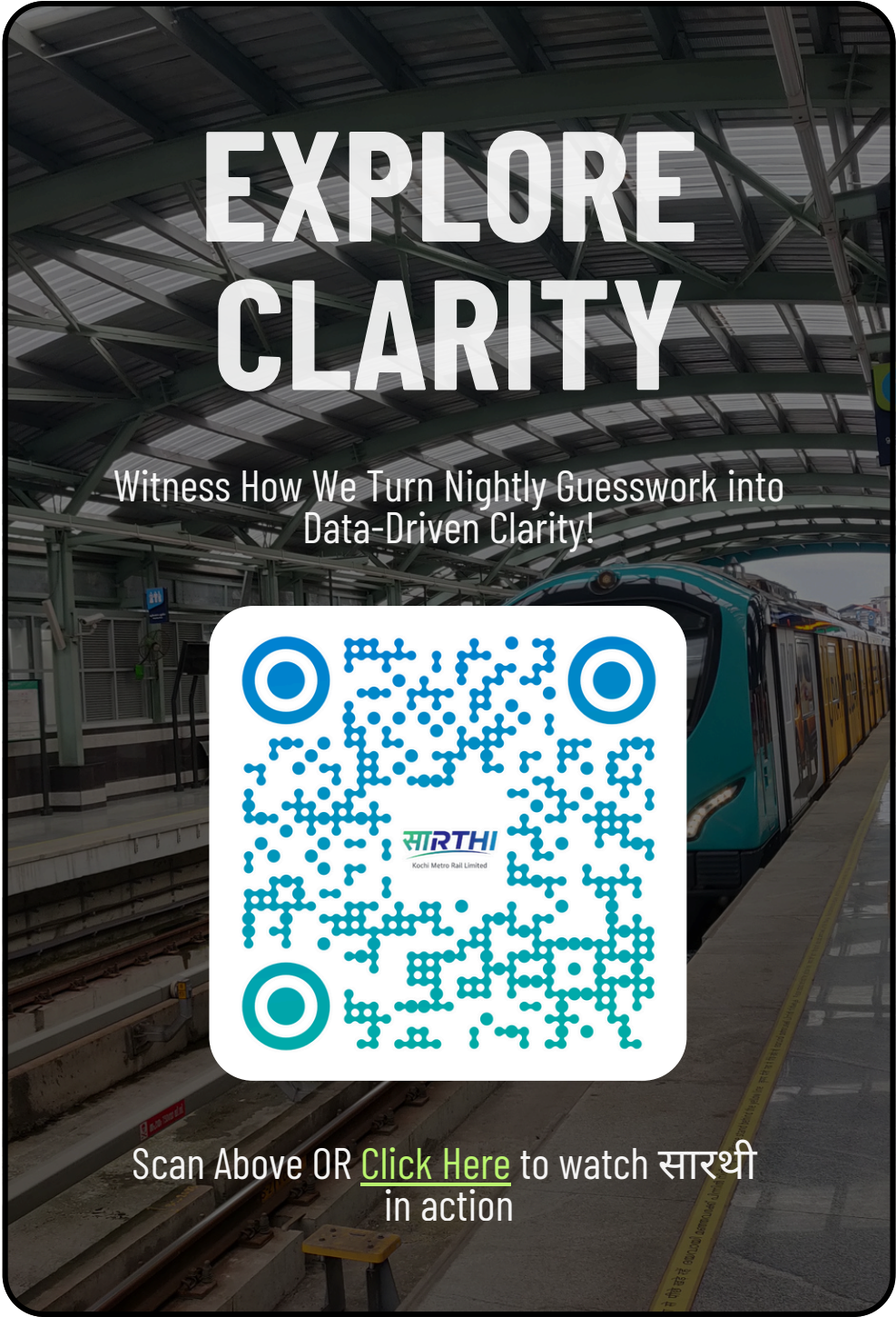
- **Meets Branding Commitments** → Avoids penalties, protects revenue
- **Builds Passenger Trust** → Ensures punctuality, improves commuter satisfaction
- **Positions Metro as Future-Ready** → Smart, scalable & aligned with modern transports



REFERENCES

- Kochi Metro Rail Limited (KMRL) – Official Website [\[Link\]](#)
- F.A. Rijksen, "Train Schedule Optimization with Machine Learning" Tilburg University, 2020 [\[Link\]](#)
- Sun et al. (2022), "AI in railway transportation", Transp. Res. Part C, 143, 103909 [\[Link\]](#)
- AI in Railways: Predictive Analytics for Maintenance [\[Link\]](#)
- Autonomous Scheduling for Public Transportation with Agentic AI [\[Link\]](#)
- Role of Large Language Models (LLM) in Decision Making [\[Link\]](#)

CONCEPT



COMPARATIVE ANALYSIS

Aspect	Toshiba TrueLine	Siemens TPS	Proposed System (Saarthi)
Data Inputs & Compatibility	✓	✓	✓
Multi-Source Integration	✗	✗	✓
AI-Driven Optimisation	✗	✗	✓
Predictive & Swarm Intelligence	✗	✗	✓
ML Feedback Loops	✗	✗	✓

[CLICK HERE to Access the Live Prototype](#)

